

01__eda__onet__oews

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1 01 — EDA: O*NET/OEWS crosswalk coverage + skill-matching sanity check

Purpose (Phase 0, last step before committing to Phase 1): 1. Confirm what % of O*NET occupations find a BLS wage match via the SOC crosswalk. 2. Sanity-check the weighted-overlap skill-matching baseline on a known skill set.

This is exploration only — nothing here is final. If the approach looks sound, the logic moves into `src/` properly for Phase 1.

```
[1]: import sys
from pathlib import Path

# Notebook runs from notebooks/, so add project root to the path
project_root = Path.cwd().parent
sys.path.insert(0, str(project_root))

import pandas as pd

from src.data.onet_loader import load_occupation_data, load_essential_skills,
    ↳load_transferable_skills, build_skill_importance_matrix,
    ↳build_full_skill_importance_matrix
from src.data.oews_loader import load_national_wages
from src.data.soc_crosswalk import add_bls_soc_column, join_onet_to_oews
from src.features.skill_matching import weighted_overlap_match
```

```
[2]: occ = load_occupation_data()
skills = load_essential_skills()
transferable = load_transferable_skills()
wages = load_national_wages()

print(f"O*NET occupations: {len(occ)}")
print(f"Essential skill ratings: {len(skills)}")
print(f"Transferable skill ratings: {len(transferable)}")
print(f"BLS OEWS occupations: {len(wages)}")
```

[oews_loader] 7 rows have suppressed wage data - kept as NaN, not dropped.

O*NET occupations: 1016

Essential skill ratings: 17880

Transferable skill ratings: 44700
BLS OEWS occupations: 1401

1.1 1. Crosswalk coverage

Every ONET-SOC code should collapse to a BLS SOC code (drop the .XX suffix). Not every one will find a wage match in OEWS — e.g. ONET includes military and some emerging/not-yet-OEWS-tracked occupations. We want to know the real %, not assume 100%.

```
[3]: occ_with_bls = add_bls_soc_column(occ)
joined = join_onet_to_oews(occ, wages)

n_total = len(occ_with_bls)
n_matched = joined["a_median"].notna().sum()
n_unmatched = n_total - n_matched

print(f"O*NET occupations: {n_total}")
print(f"Found a BLS wage match: {n_matched} ({n_matched / n_total:.1%})")
print(f"No BLS wage match: {n_unmatched} ({n_unmatched / n_total:.1%})")
```

O*NET occupations: 1016
Found a BLS wage match: 956 (94.1%)
No BLS wage match: 60 (5.9%)

```
[4]: # Look at a sample of unmatched occupations to understand WHY they don't match
# (expected: military-specific, or O*NET codes with no OEWS civilian
# ↪ counterpart)
unmatched = joined[joined["a_median"].isna()][["onetsoc_code", "title"]]
unmatched.head(20)
```

```
[4]:
```

	onetsoc_code	title
3	11-1031.00	Legislators
60	13-1021.00	Buyers and Purchasing Agents, Farm Products
61	13-1022.00	Wholesale and Retail Buyers, Except Farm Products
62	13-1023.00	Purchasing Agents, Except Wholesale, Retail, a...
93	13-2022.00	Appraisers of Personal and Business Property
94	13-2023.00	Appraisers and Assessors of Real Estate
272	21-1011.00	Substance Abuse and Behavioral Disorder Counse...
275	21-1014.00	Mental Health Counselors
343	25-2055.00	Special Education Teachers, Kindergarten
344	25-2056.00	Special Education Teachers, Elementary School
361	25-9042.00	Teaching Assistants, Preschool, Elementary, Mi...
362	25-9043.00	Teaching Assistants, Special Education
364	25-9049.00	Teaching Assistants, All Other
379	27-2011.00	Actors
387	27-2031.00	Dancers
390	27-2042.00	Musicians and Singers
391	27-2091.00	Disc Jockeys, Except Radio

392	27-2099.00	Entertainers and Performers, Sports and Relate...
472	29-2011.00	Medical and Clinical Laboratory Technologists
473	29-2011.01	Cytogenetic Technologists

1.2 2. Skill-matching baseline sanity check

Build the occupation x skill importance matrix, then test with a known skill profile. If this surfaces plausible occupations near the top, the baseline approach is worth building on. If it produces nonsense, we need to rethink before Phase 1 — better to find that out now on a handful of manual checks than after building the full pipeline on top of it.

```
[5]: importance_matrix = build_full_skill_importance_matrix(skills, transferable)
print(f"Matrix shape: {importance_matrix.shape} (occupations x skills)")
print(f"Skill elements available ({len(importance_matrix.columns)} total):
↳{list(importance_matrix.columns)}")

# Compare against Essential Skills alone, to see how much the 894-occupation gap
# from Phase 0 closes once Transferable Skills (Cross-Functional Skills) is
↳included.
essential_only_matrix = build_skill_importance_matrix(skills)
print(f"\nFor comparison - Essential Skills alone: {essential_only_matrix.
↳shape} (was missing Cross-Functional Skills like Programming, Negotiation,
↳Coordination)")
```

```
Matrix shape: (894, 35) (occupations x skills)
Skill elements available (35 total): ['Active Learning', 'Active Listening',
'Complex Problem Solving', 'Coordination', 'Critical Thinking', 'Equipment
Maintenance', 'Equipment Selection', 'Installation', 'Instructing', 'Judgment
and Decision Making', 'Learning Strategies', 'Management of Financial
Resources', 'Management of Material Resources', 'Management of Personnel
Resources', 'Mathematics', 'Monitoring', 'Negotiation', 'Operation and Control',
'Operations Analysis', 'Operations Monitoring', 'Persuasion', 'Programming',
'Quality Control Analysis', 'Reading Comprehension', 'Repairing', 'Science',
'Service Orientation', 'Social Perceptiveness', 'Speaking', 'Systems Analysis',
'Systems Evaluation', 'Technology Design', 'Time Management', 'Troubleshooting',
'Writing']
```

```
For comparison - Essential Skills alone: (894, 10) (was missing Cross-Functional
Skills like Programming, Negotiation, Coordination)
```

1.3 2b. Investigating the 894 vs. 1,016 occupation gap

The combined matrix still covers only 894 of 1,016 occupations — identical to Essential Skills alone, so adding Transferable Skills didn't close this gap. That means whatever is excluding ~122 occupations affects BOTH files identically. Two candidate explanations to distinguish between:

1. These occupations simply don't appear in either raw skills file at all (e.g. newer/emerging O*NET-SOC codes not yet skill-surveyed).
2. They appear in the raw files, but only under the LV (Level) scale, not IM (Importance) — since `build_skill_importance_matrix` filters to `scale_id`

== "IM" only, an occupation with LV ratings but no IM ratings would silently disappear at the pivot step.

```
[6]: all_occ_codes = set(occ["onetsoc_code"])
matrix_occ_codes = set(importance_matrix.index)
missing_from_matrix = all_occ_codes - matrix_occ_codes
print(f"Occupations missing from the combined skill matrix:␣
↳{len(missing_from_matrix)}")

combined_skills = pd.concat([skills, transferable], ignore_index=True)
missing_in_row_entirely = missing_from_matrix -␣
↳set(combined_skills["onetsoc_code"])
present_in_row_no_im = missing_from_matrix &␣
↳set(combined_skills["onetsoc_code"])

print(f" ...absent from raw skill files entirely:␣
↳{len(missing_in_row_entirely)}")
print(f" ...present in raw files but excluded at the IM-scale pivot (likely␣
↳LV-only): {len(present_in_row_no_im)}")

missing_titles = occ[occ["onetsoc_code"].
↳isin(missing_from_matrix)][["onetsoc_code", "title"]]
print(f"\nSample of missing occupations:")
missing_titles.head(20)
```

Occupations missing from the combined skill matrix: 122

...absent from raw skill files entirely: 122

...present in raw files but excluded at the IM-scale pivot (likely LV-only): 0

Sample of missing occupations:

```
[6]:      onetsoc_code      title
3      11-1031.00      Legislators
7      11-2032.00      Public Relations Managers
33     11-9039.00      Education Administrators, All Other
49     11-9179.00      Personal Service Managers, All Other
52     11-9199.00      Managers, All Other
79     13-1082.00      Project Management Specialists
87     13-1199.00      Business Operations Specialists, All Other
97     13-2051.00      Financial and Investment Analysts
100    13-2054.00      Financial Risk Specialists
106    13-2099.00      Financial Specialists, All Other
125    15-1255.00      Web and Digital Interface Designers
127    15-1299.00      Computer Occupations, All Other
131    15-1299.04      Penetration Testers
133    15-1299.06      Digital Forensics Analysts
134    15-1299.07      Blockchain Engineers
142    15-2051.00      Data Scientists
```

```

145 15-2099.00 Mathematical Science Occupations, All Other
178 17-2199.00 Engineers, All Other
190 17-3019.00 Drafters, All Other
201 17-3028.00 Calibration Technologists and Technicians

```

```

[7]: # If any occupations ARE present in raw files but missing IM ratings, look at
# what scale_id values they DO have - confirms/refutes hypothesis 2 directly.
if len(present_in_raw_no_im) > 0:
    sample_code = next(iter(present_in_raw_no_im))
    sample_rows = combined_skills[combined_skills["onetsoc_code"] ==
↪sample_code]
    print(f"Sample occupation with raw data but no IM scale: {sample_code}")
    print(sample_rows[["element_name", "scale_id", "data_value"]])
else:
    print("No occupations fall into the 'present but no IM scale' case - "
          "the gap is entirely occupations absent from the raw skill files.")

```

No occupations fall into the 'present but no IM scale' case - the gap is entirely occupations absent from the raw skill files.

```

[8]: # Test case 1: a data-science-learning skill set.
# Skill names must match O*NET's 34 standard elements exactly (see printed list
↪above).
test_skills_1 = {
    "Programming": 5,
    "Mathematics": 5,
    "Complex Problem Solving": 4,
    "Critical Thinking": 4,
    "Systems Analysis": 3,
}

results_1 = weighted_overlap_match(test_skills_1, importance_matrix, occ,
↪top_n=10)
for r in results_1:
    print(f"{r.score:.3f} {r.title} ({r.onetsoc_code})")

```

```

0.989 Computer Numerically Controlled Tool Programmers (51-9162.00)
0.988 Biostatisticians (15-2041.01)
0.987 Statistical Assistants (43-9111.00)
0.986 Physicists (19-2012.00)
0.983 Computer Programmers (15-1251.00)
0.982 Bioinformatics Technicians (15-2099.01)
0.982 Biologists (19-1029.04)
0.981 Clinical Data Managers (15-2051.02)
0.981 Statisticians (15-2041.00)
0.980 Geographic Information Systems Technologists and Technicians
(15-1299.02)

```

```
[9]: # Test case 2: a people-management-leaning skill set, as a contrast check -
# should surface clearly different occupations than test 1.
```

```
test_skills_2 = {
    "Management of Personnel Resources": 5,
    "Coordination": 5,
    "Negotiation": 4,
    "Persuasion": 4,
    "Speaking": 4,
}

results_2 = weighted_overlap_match(test_skills_2, importance_matrix, occ,
    ↪top_n=10)
for r in results_2:
    print(f"{r.score:.3f} {r.title} ({r.onetsoc_code})")
```

```
0.998 Social and Community Service Managers (11-9151.00)
0.997 Solar Energy Installation Managers (47-1011.03)
0.997 Biofuels Production Managers (11-3051.03)
0.997 First-Line Supervisors of Housekeeping and Janitorial Workers
(37-1011.00)
0.997 Manufactured Building and Mobile Home Installers (49-9095.00)
0.997 Information Technology Project Managers (15-1299.09)
0.997 First-Line Supervisors of Firefighting and Prevention Workers
(33-1021.00)
0.997 First-Line Supervisors of Helpers, Laborers, and Material Movers, Hand
(53-1042.00)
0.996 First-Line Supervisors of Mechanics, Installers, and Repairers
(49-1011.00)
0.996 First-Line Supervisors of Material-Moving Machine and Vehicle Operators
(53-1043.00)
```

1.4 3. Verdict

- **Crosswalk coverage: 94.1% matched (956/1,016).** Acceptable — the unmatched 60 don't share one obvious cause (a mix of legislative, some postsecondary- teacher subtypes, performing-arts, and a couple of lab-tech specialties); documented as a known limitation rather than something to chase further right now.
- **Test 1 (quant skills): clean and plausible** — Biostatisticians, Statisticians, Computer Programmers, Physicists, Bioinformatics Technicians, GIS Technologists all surface near the top, using the full 35-skill space with no partial-overlap warnings.
- **Test 2 (management skills): clean and plausible after the Transferable Skills fix** — Social and Community Service Managers, IT Project Managers, and a cluster of First-Line Supervisor roles now surface, replacing the earlier Postsecondary-Teacher/Clergy/Lawyer result that turned out to be an artifact of matching on only 1-2 of the 5 specified skills (see section 2b and skill_matching.py's fix note).
- **894 vs. 1,016 gap: see section 2b's diagnostic output above** for whether it's occupations missing from the raw files entirely, or present but excluded at the IM-scale filter.

Verdict: proceed to Phase 1. The matching approach is sound on the full skill space. Known, documented limitations: ~5.9% crosswalk gap, and whatever section 2b's diagnostic reveals about the 894-occupation subset — both go in the README's Limitations section rather than blocking further work.